

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – MODEL COMPARISON PRESENTATION (BERT VS GPT VS T5 VS LLAMA)

Course Code:	CSA6503	Course Title:	Generative AI and Large Language Models
CO Assessed:	CO4 – Precision (BL3, BL4,BL5)	Assessment:	Assessment Tool 2 - Model Comparison Presentation (BERT vs GPT vs T5 vs LLaMA)
Weightage:	10% of CIA	Total Marks:	2 * 50= 100
CO4 Statement:	<i>Design and implement multimodal Generative AI applications integrating text, image, and speech models for engineering applications.(BL3, BL4,BL5)</i>		
Student Name:	M Prabhas	Reg. No.:	192472064
Section / Batch:	Slot D/2024	Date:	24/08/2024

Instructions to Students

- **Answer any 2 questions. Each question carries 50 marks. Total: 100 marks.**
- **Compare all four models: BERT, GPT, T5, and LLaMA.**
- Explain the **architecture and working principle** of each model.
- Compare their **pre-training objectives and capabilities**.
- Map each model to appropriate **real-world applications**.
- Identify the **strengths and limitations** of each model.
- Support the presentation using **credible research papers, official documentation, benchmarks, or other reliable sources**.
- Include at least one **comparison table or diagram**.
- Use relevant **real-world case studies or application examples**.
- Conclude by identifying the **most suitable model for the selected application** and justify the decision.
- Include proper **references/citations** in the presentation.

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

SECTION A : MODEL COMPARISON PRESENTATION (BERT VS GPT VS T5 VS LLAMA)

ARCHITECTURE-BASED COMPARISON

1. Compare the architectures of BERT, GPT, T5, and LLaMA. Explain the major architectural differences and identify the type of NLP tasks each architecture is best suited for.
2. Compare encoder-only, decoder-only, and encoder-decoder architectures using BERT, GPT, and T5 as representative models. Explain where LLaMA fits and justify your classification.
3. Compare the Transformer architecture implementation in BERT, GPT, T5, and LLaMA. Discuss the role of self-attention, positional information, and feed-forward layers.
4. Compare how BERT, GPT, T5, and LLaMA process input sequences and generate outputs. Illustrate the processing flow of each model.
5. Compare the pre-training objectives of BERT, GPT, T5, and LLaMA. Explain how the objective influences the capabilities of each model.
6. Compare Masked Language Modeling, Causal Language Modeling, and Text-to-Text learning using BERT, GPT, and T5. Explain how LLaMA's training approach differs.
7. Compare the attention mechanisms and contextual understanding capabilities of BERT, GPT, T5, and LLaMA. Discuss their suitability for different NLP applications.
8. Compare the input and output representations of BERT, GPT, T5, and LLaMA. Explain how these differences affect their applications.
9. Compare the architectural strengths and limitations of BERT, GPT, T5, and LLaMA for language understanding and language generation.
10. Prepare a detailed comparison of BERT, GPT, T5, and LLaMA based on architecture, training objective, input/output structure, and major capabilities.

APPLICATION MAPPING

11. Compare BERT, GPT, T5, and LLaMA for text classification. Identify the most suitable model and justify your choice with real-world examples.
12. Compare BERT, GPT, T5, and LLaMA for text summarization. Explain which model or architecture is most appropriate and why.
13. Compare BERT, GPT, T5, and LLaMA for machine translation. Discuss their strengths, limitations, and suitable use cases.
14. Compare the four models for question answering systems. Identify which architecture is suitable for extractive and generative question answering.
15. Compare BERT, GPT, T5, and LLaMA for AI chatbot development. Map each model to suitable chatbot applications.
16. Compare the four models for code generation and code understanding. Discuss which model is more appropriate for each task.
17. Compare BERT, GPT, T5, and LLaMA for sentiment analysis and other text classification tasks. Justify the best model for the application.

18. Compare the four models for information extraction, including named entity recognition and relation extraction. Explain the advantages and limitations of each.
19. Compare BERT, GPT, T5, and LLaMA for document-based question answering and RAG applications. Identify the role each model can play in a RAG pipeline.
20. Compare the four models for multilingual NLP applications. Discuss their suitability for applications involving English and Indian languages.

PERFORMANCE AND PRACTICAL COMPARISON

21. Compare BERT, GPT, T5, and LLaMA based on model size, computational requirements, training requirements, and inference requirements.
22. Compare the four models in terms of fine-tuning and transfer learning. Explain which models are more suitable for task-specific adaptation.
23. Compare BERT, GPT, T5, and LLaMA based on their suitability for resource-constrained environments. Discuss factors such as memory, computation, and inference cost.
24. Compare the four models based on training data requirements and pre-training strategies. Explain how these factors influence model performance.
25. Compare BERT, GPT, T5, and LLaMA for zero-shot, one-shot, and few-shot learning. Explain how prompting and fine-tuning influence their performance.
26. Compare the four models based on their ability to handle long-context inputs. Discuss the challenges and possible applications.
27. Compare BERT, GPT, T5, and LLaMA in terms of model adaptability and customization. Discuss fine-tuning, parameter-efficient approaches, and prompting.
28. Compare the four models based on open-source availability, accessibility, customization, and deployment possibilities. Discuss the practical implications for developers.
29. Compare BERT, GPT, T5, and LLaMA from the perspective of accuracy, computational efficiency, scalability, and application suitability.
30. Conduct a comparative study of BERT, GPT, T5, and LLaMA and determine which model would be most appropriate for a college-level AI application. Justify your recommendation.

REAL-WORLD CASE STUDY COMPARISON

31. A university wants to develop an AI student-support system. Compare BERT, GPT, T5, and LLaMA and recommend the most suitable model for the system.
32. A company wants to develop an automated customer-support chatbot. Compare the four models and determine which architecture would provide the most suitable solution.
33. A news organization wants to automatically summarize thousands of news articles. Compare BERT, GPT, T5, and LLaMA and recommend an appropriate model.
34. A multinational organization wants to develop a multilingual translation system. Compare BERT, GPT, T5, and LLaMA and identify the most appropriate architecture.
35. A software company wants to develop an AI coding assistant. Compare the four models based on code understanding, generation, context handling, and deployment requirements.
36. A college wants to build a RAG-based document question-answering system using academic regulations and course documents. Compare BERT, GPT, T5, and LLaMA and recommend a suitable model.
37. A healthcare organization wants to build an NLP system for classifying and summarizing medical documents. Compare BERT, GPT, T5, and LLaMA and discuss the suitability and limitations of each.
38. An e-commerce company wants to develop an AI system capable of customer sentiment analysis, product review summarization, and chatbot support. Compare BERT, GPT, T5, and LLaMA and recommend appropriate models for each task.
39. A startup has limited computational resources but wants to develop an AI-powered educational assistant. Compare BERT, GPT, T5, and LLaMA in terms of capability, resource requirements, customization, and deployment.

40. A company wants to develop a general-purpose Generative AI application supporting text generation, summarization, question answering, translation, and chatbot interaction. Conduct a comprehensive comparison of BERT, GPT, T5, and LLaMA and recommend the most suitable model or combination of models.

Recommended Presentation Structure: BERT vs GPT vs T5 vs LLaMA

Slide 1 – Title

BERT vs GPT vs T5 vs LLaMA

Architecture-Based, Application, Performance & Real-World Comparison

Presented by: *Your Name*

Department: CSE (AI)

College: *Your College Name*

Slide 2 – Introduction

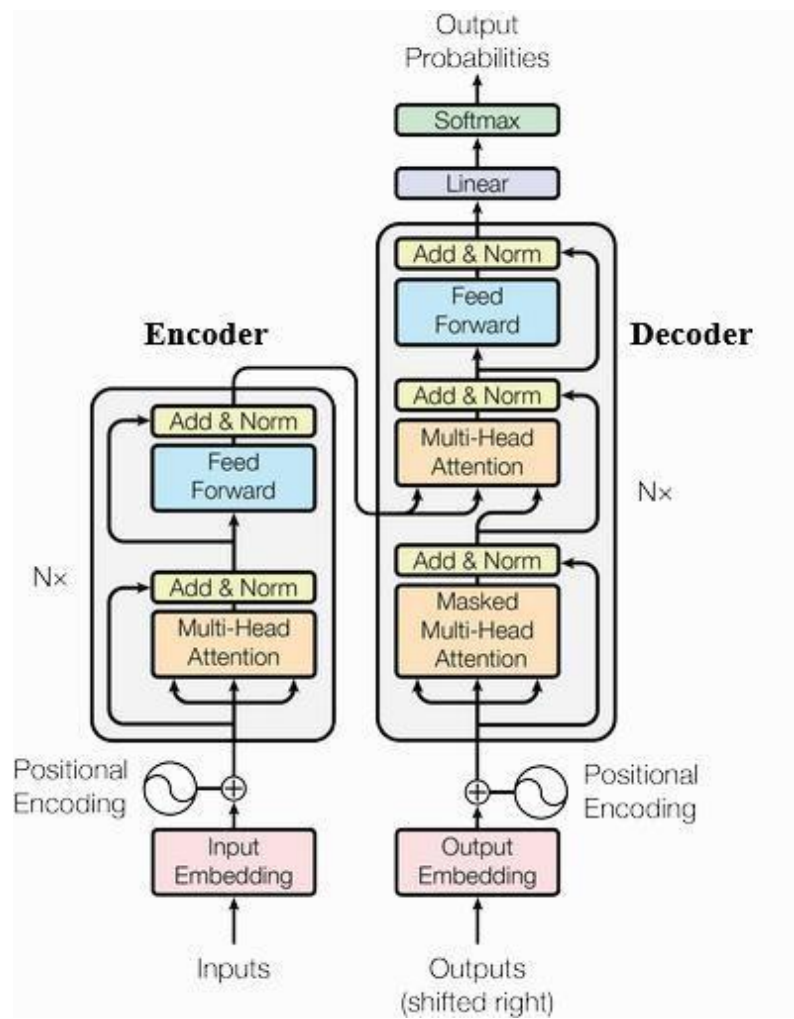
Modern NLP models are based mainly on the **Transformer architecture**. BERT, GPT, T5, and LLaMA are important Transformer-based models, but they are designed differently.

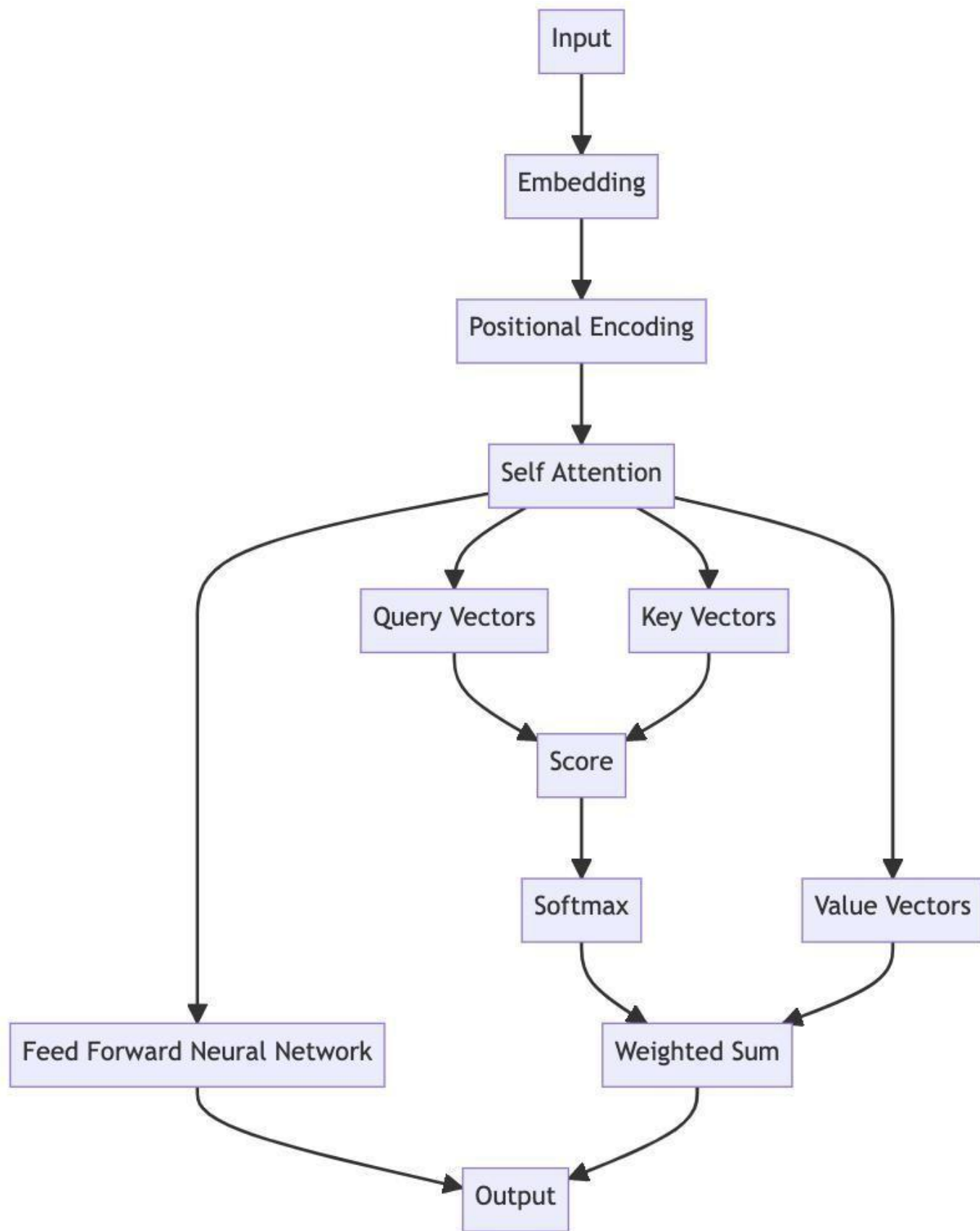
- **BERT** → Encoder-only
- **GPT** → Decoder-only
- **T5** → Encoder-decoder
- **LLaMA** → Decoder-only

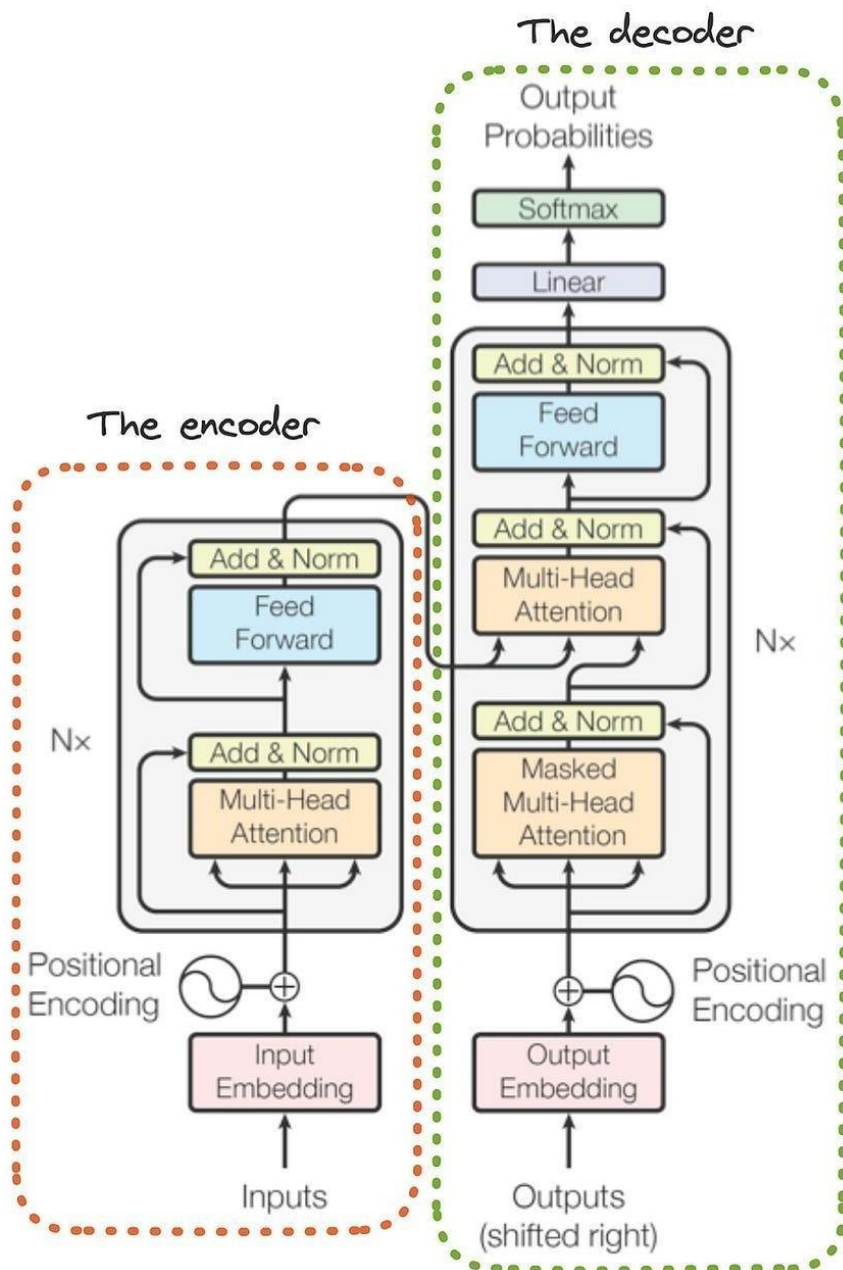
Their architectural differences affect:

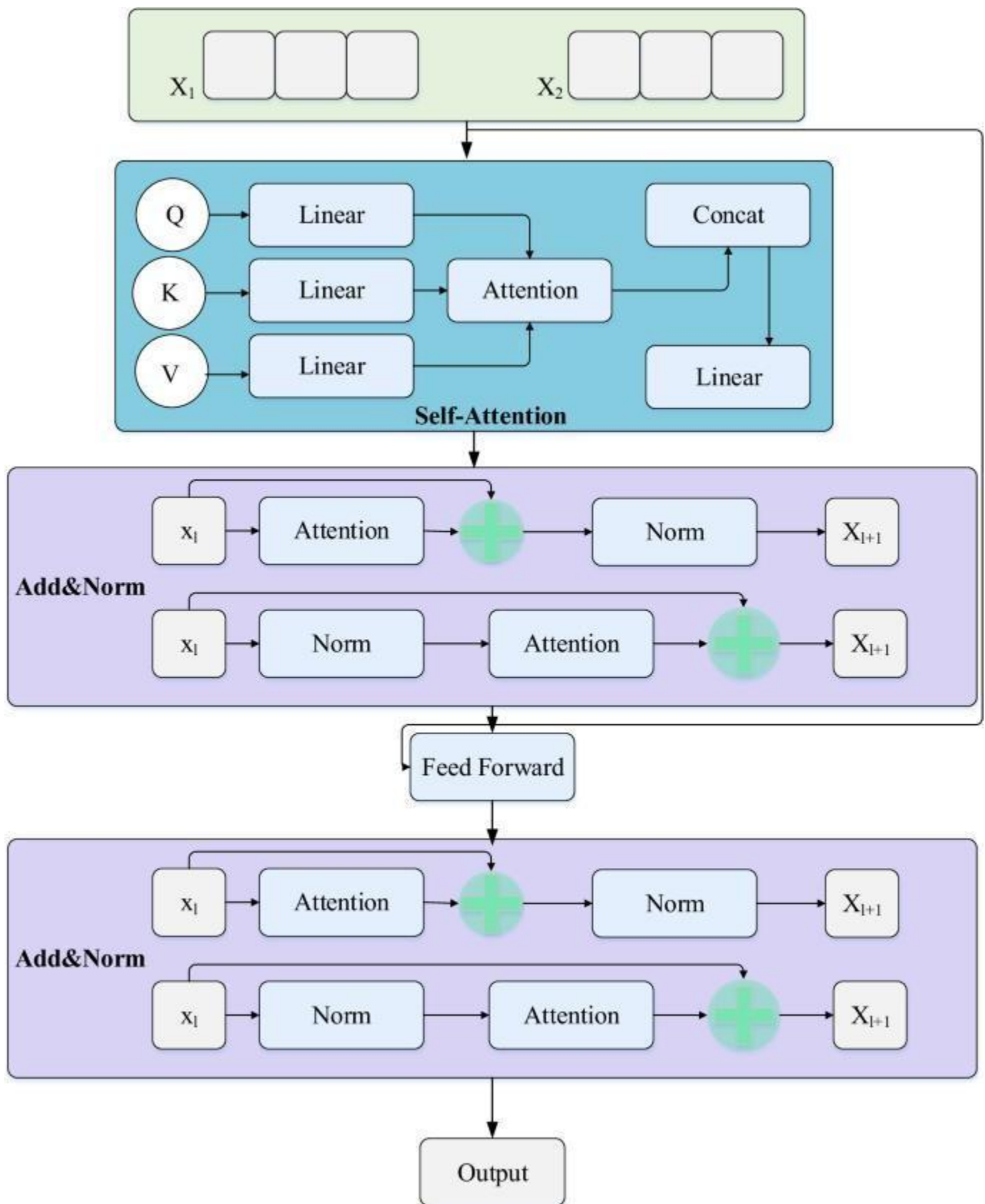
- How they understand text
 - How they generate text
 - Training objectives
 - Computational requirements
 - Suitable NLP applications
-

Slide 3 – Basic Transformer Architecture









The Transformer mainly consists of:

1. **Self-Attention**
2. **Feed-Forward Neural Network**
3. **Residual Connections**
4. **Layer Normalization**
5. **Positional Information**

Self-Attention

Self-attention allows the model to determine which words are important to understand another word.

Example:

"The student submitted the assignment because **she** completed it."

Attention helps identify what "**she**" refers to.

Feed-Forward Layer

After attention, information is processed through a feed-forward neural network to learn complex patterns.

Slide 4 – Architecture Comparison

Model	Architecture	Main Direction	Generation
BERT	Encoder-only	Bidirectional	No direct text generation
GPT	Decoder-only	Left-to-right	Yes
T5	Encoder-decoder	Bidirectional + autoregressive	Yes
LLaMA	Decoder-only	Left-to-right	Yes

Simple classification

BERT

Input → Encoder → Representation → Classification

GPT

Input → Decoder → Next Token → Next Token → ...

T5

Input → Encoder → Decoder → Output

LLaMA

Prompt → Decoder → Generated Tokens

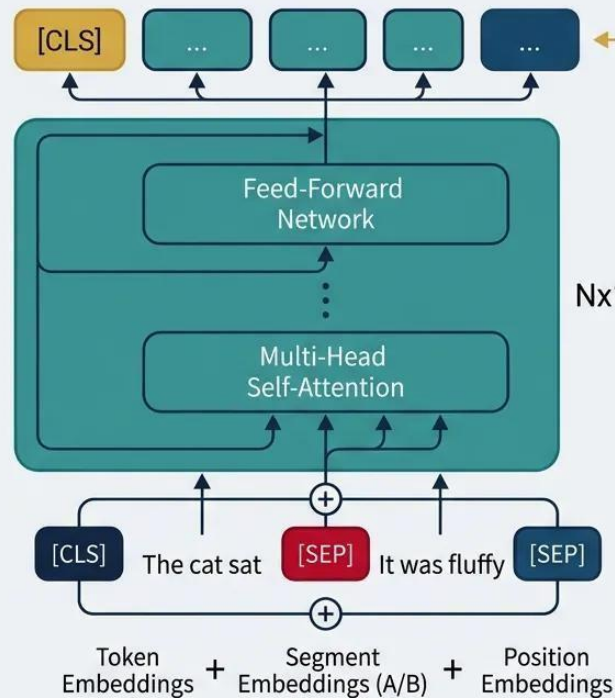
Slide 5 – BERT

BERT ARCHITECTURE

OUTPUT

TRANSFORMER
ENCODER STACK

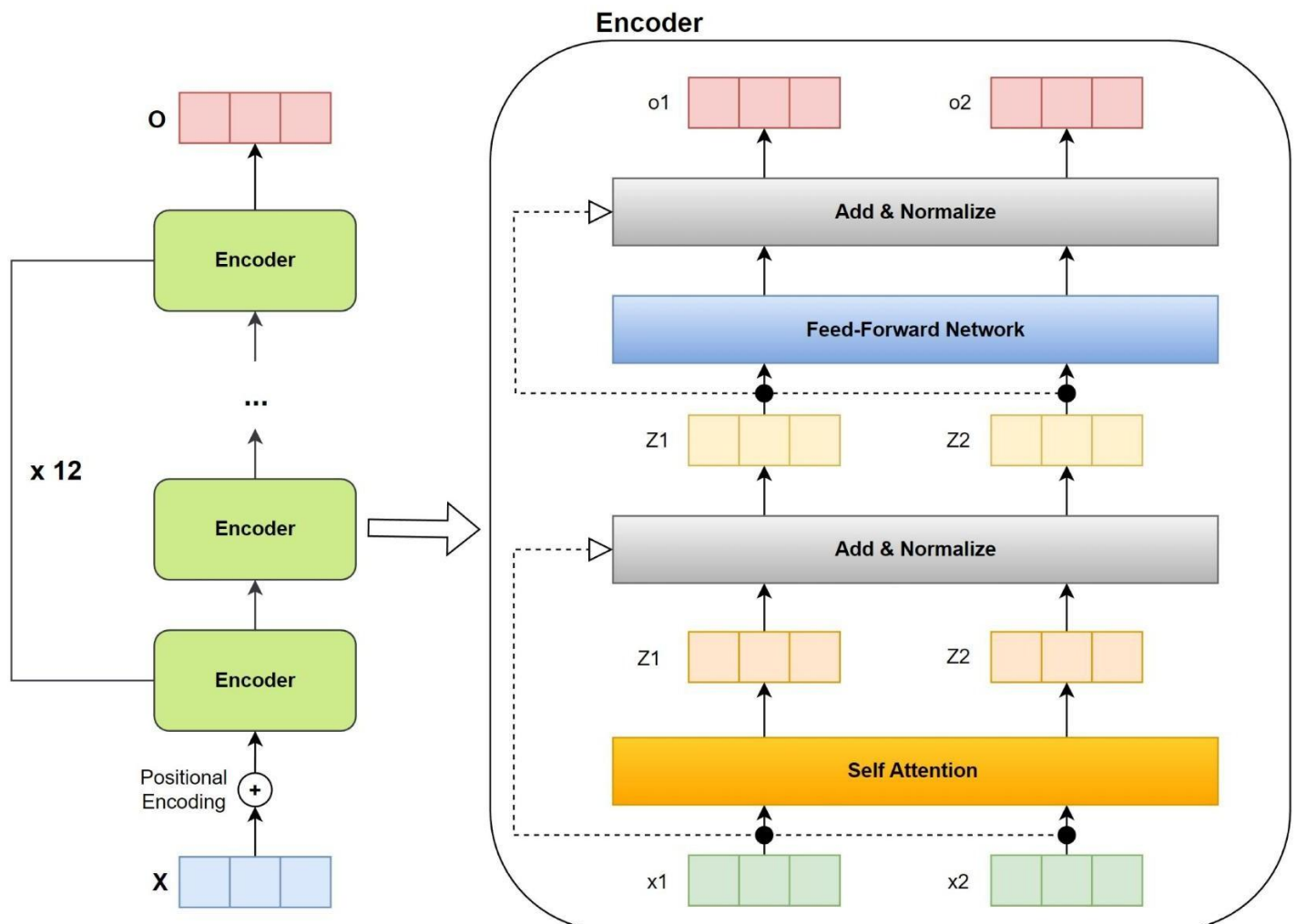
INPUT

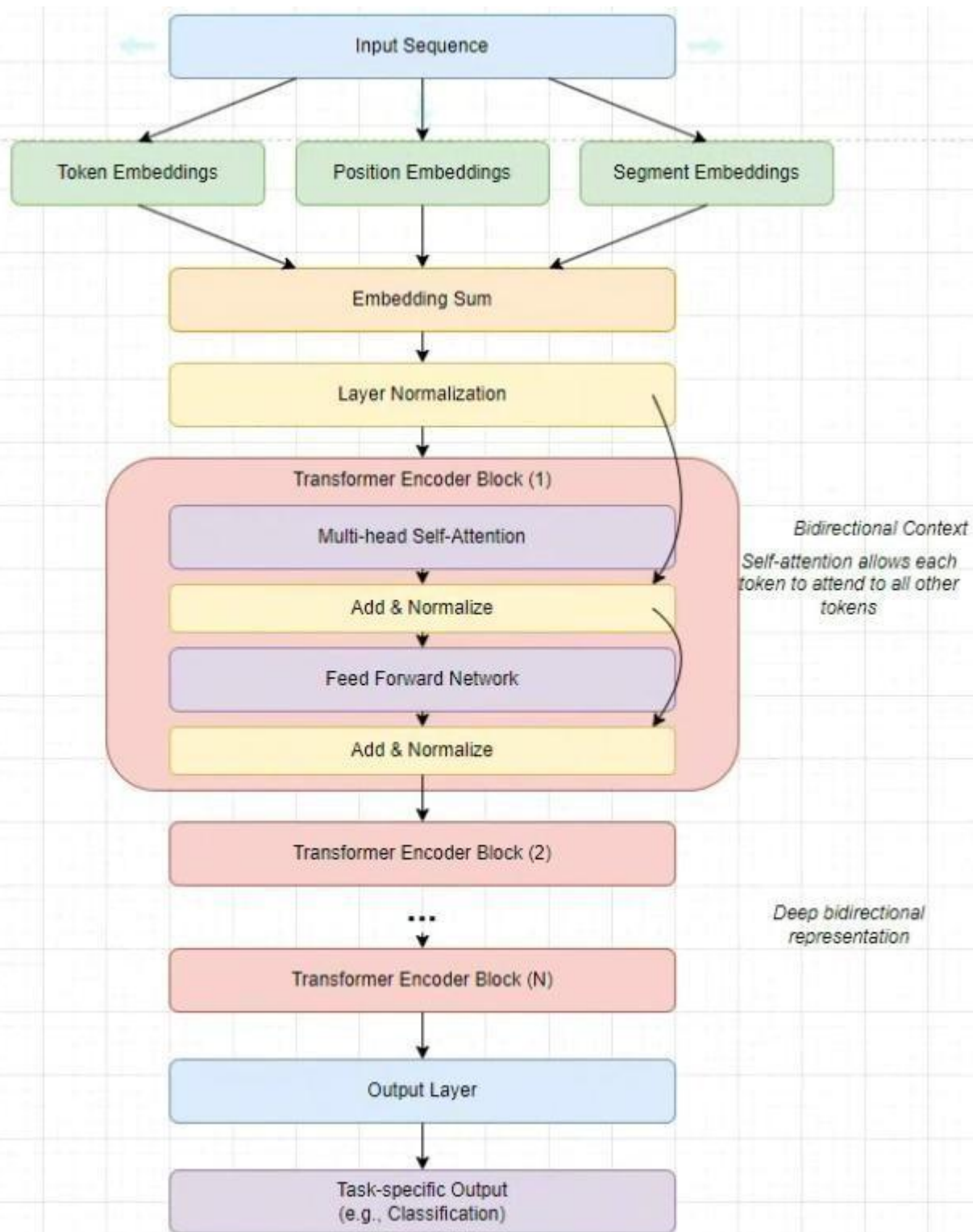


Used for classification tasks
(the pooled representation)

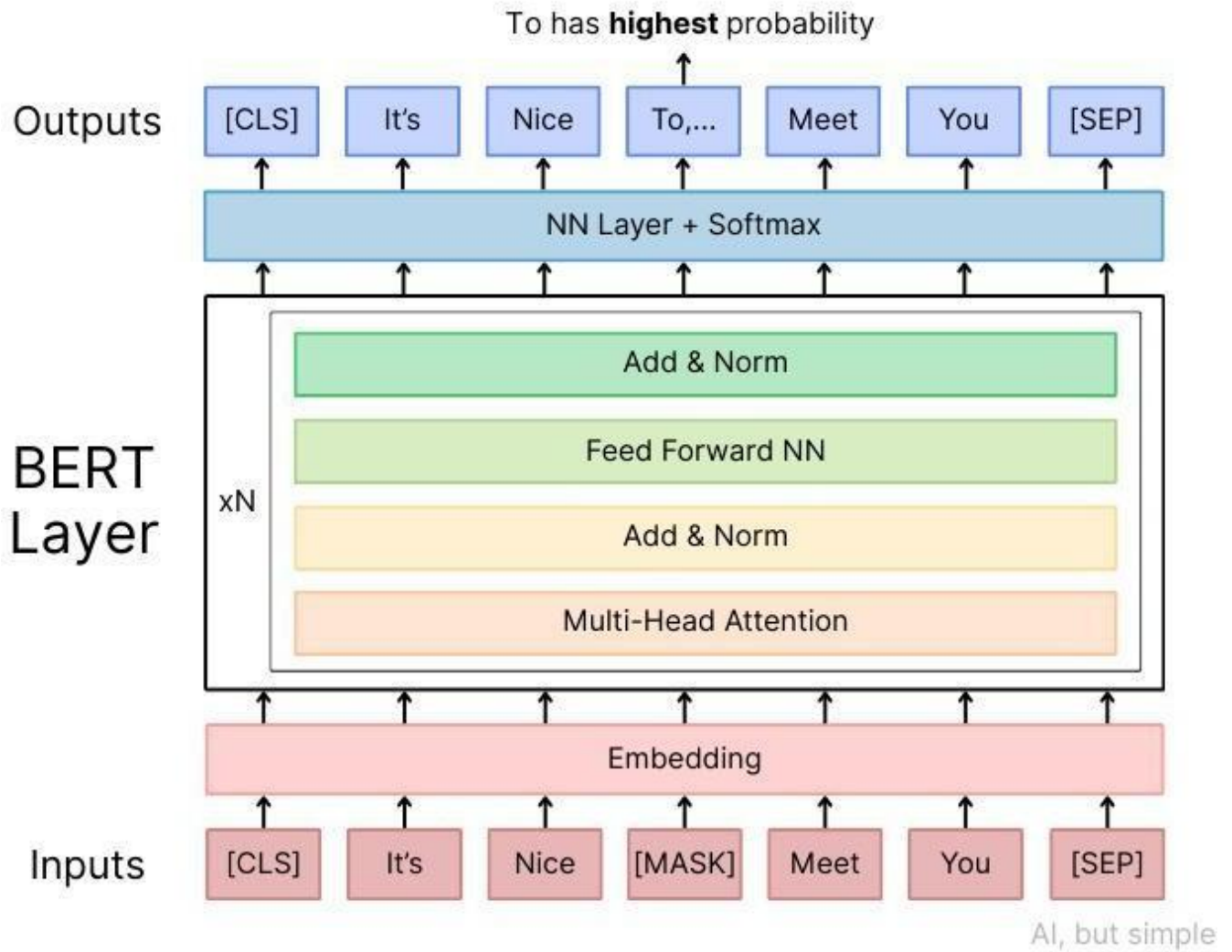
	BERT-Base	BERT-Large
Nx12	12 layers	24
	hidden units	1024
	12 attention heads	16
	110M	340M

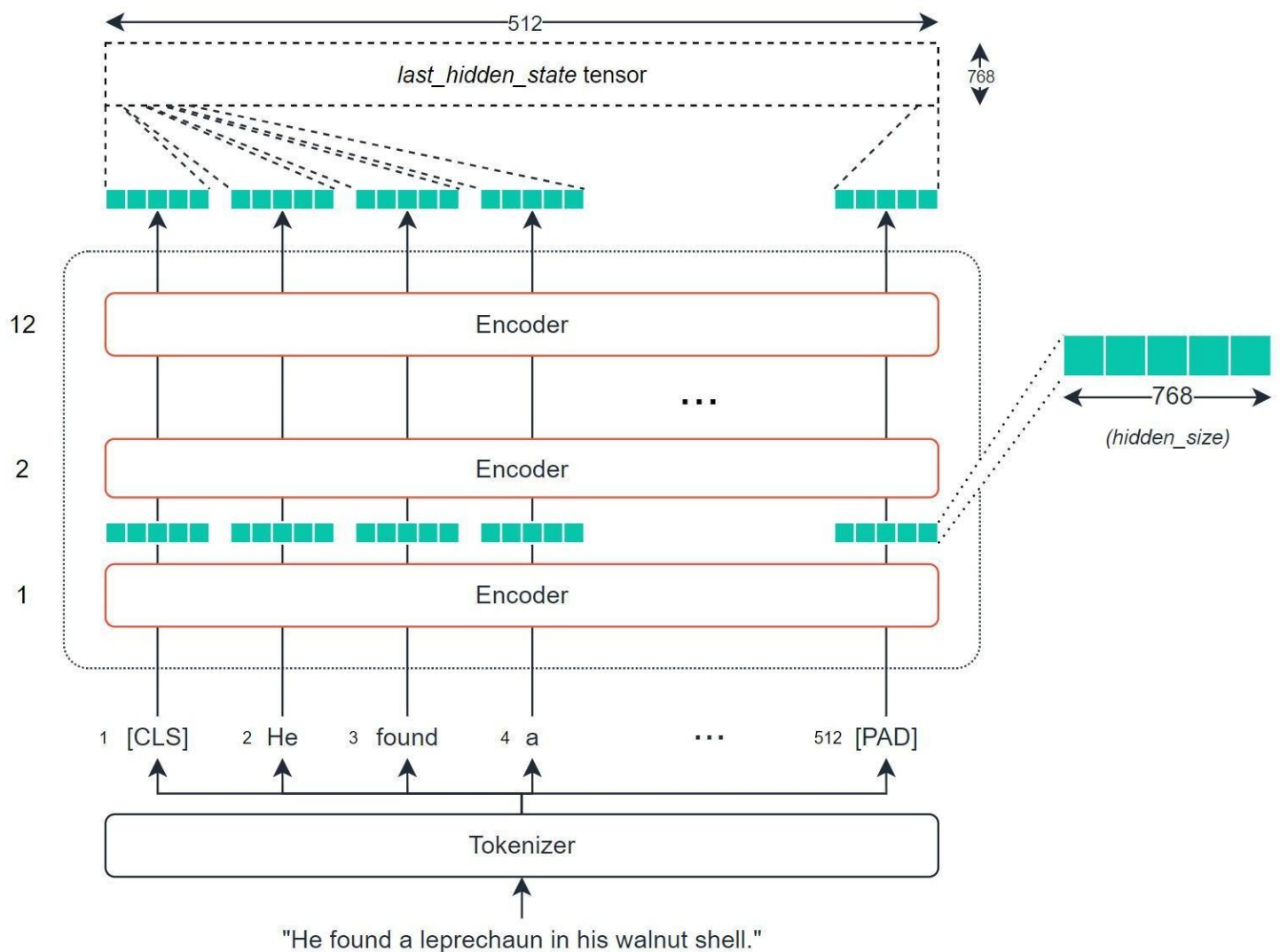
BERT uses bidirectional context through masked language modeling, producing rich contextualized representations.





BERT Language Model





BERT = Bidirectional Encoder Representations from Transformers

Architecture: Encoder-only

BERT reads the input considering context from **both left and right sides**.

Example:

"The bank is near the river."

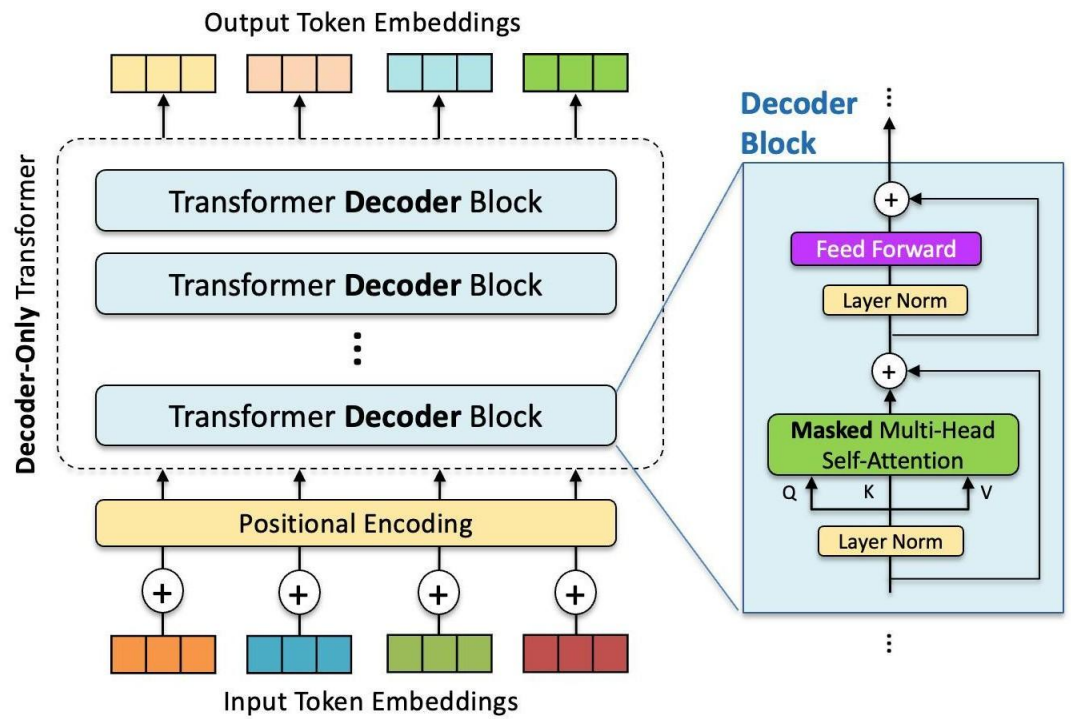
BERT can use both surrounding words to understand that "bank" refers to a riverside bank.

Best suited for

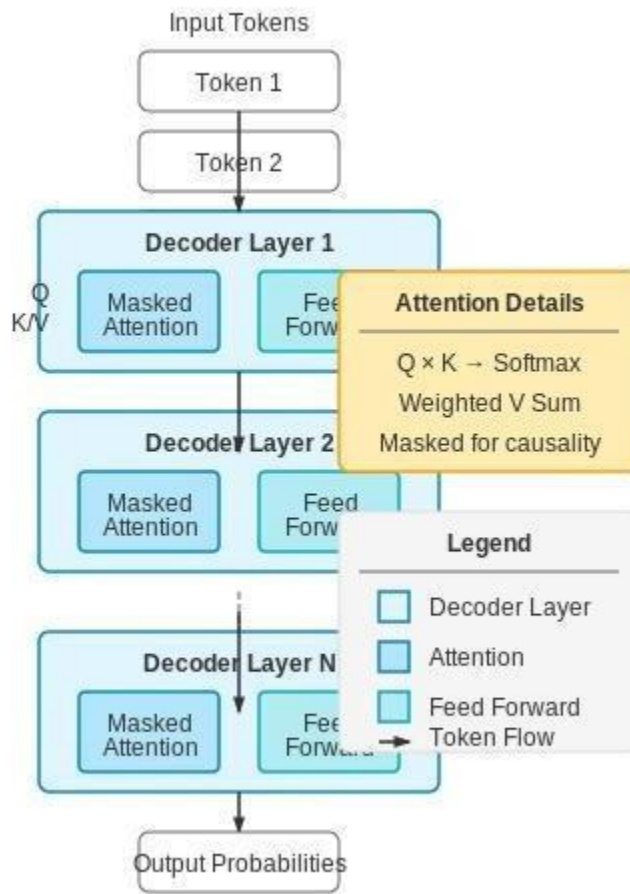
- Text classification
- Sentiment analysis
- Named Entity Recognition
- Extractive Question Answering
- Sentence similarity
- Information extraction

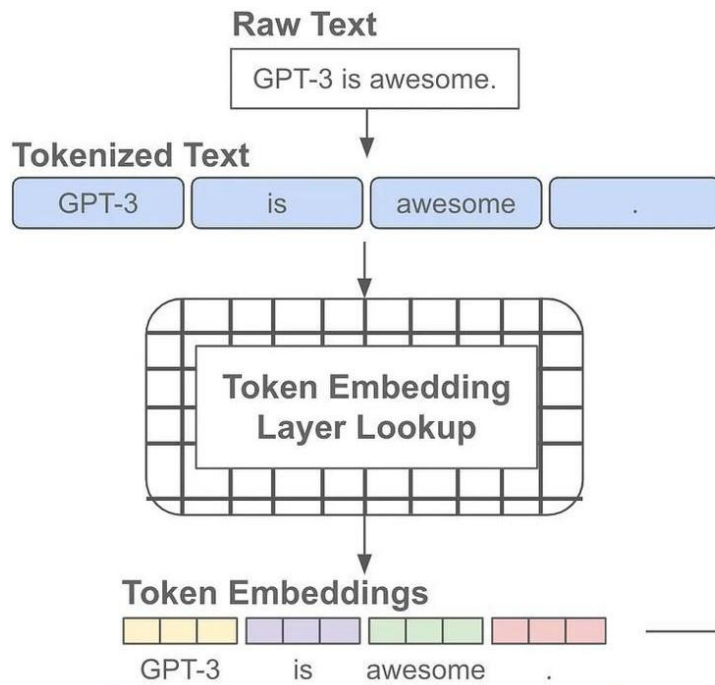
Main limitation

BERT is primarily designed for **understanding**, not free-form text generation.



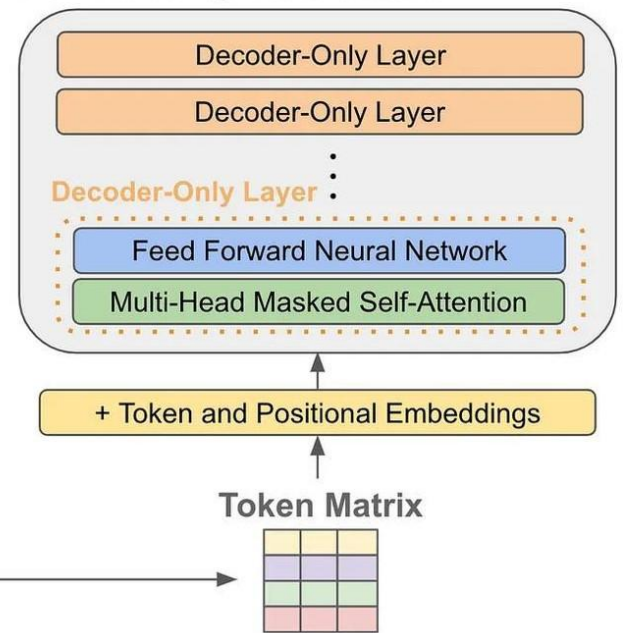
GPT Decoder Stack Architecture

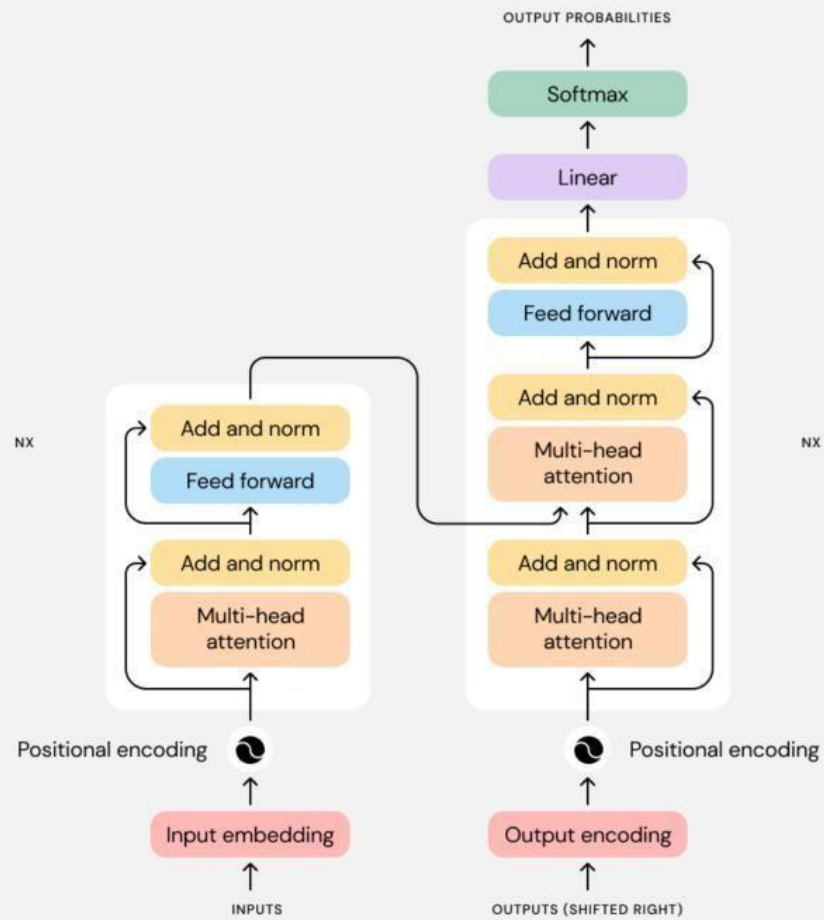




Maximum size of this sequence is determined by the context window

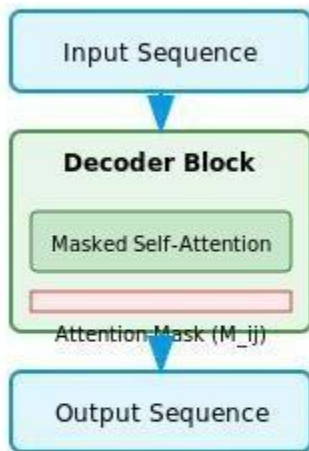
Decoder-Only Transformer



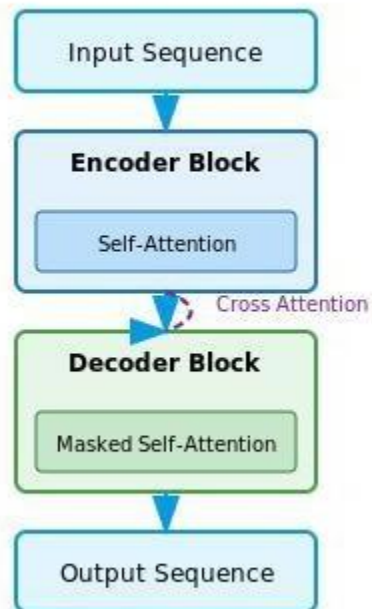


GPT Decoder-Only vs Traditional Encoder-Decoder

Decoder-Only (GPT)



Encoder-Decoder (Traditional)



GPT = Generative Pre-trained Transformer

Architecture: Decoder-only

GPT processes text from **left to right**.

Example:

"Artificial intelligence is..."

The model predicts:

"changing"

Then:

"Artificial intelligence is changing..."

and predicts the next token.

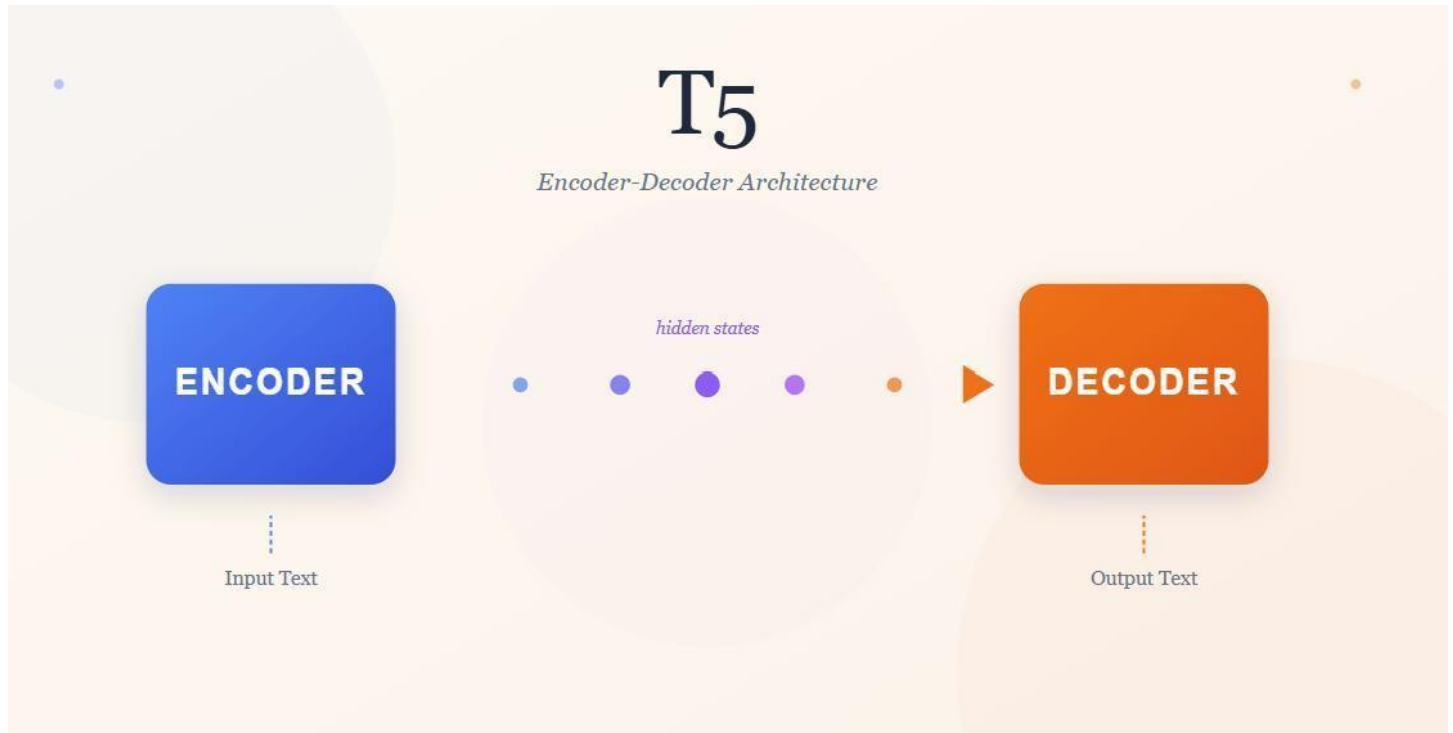
Best suited for

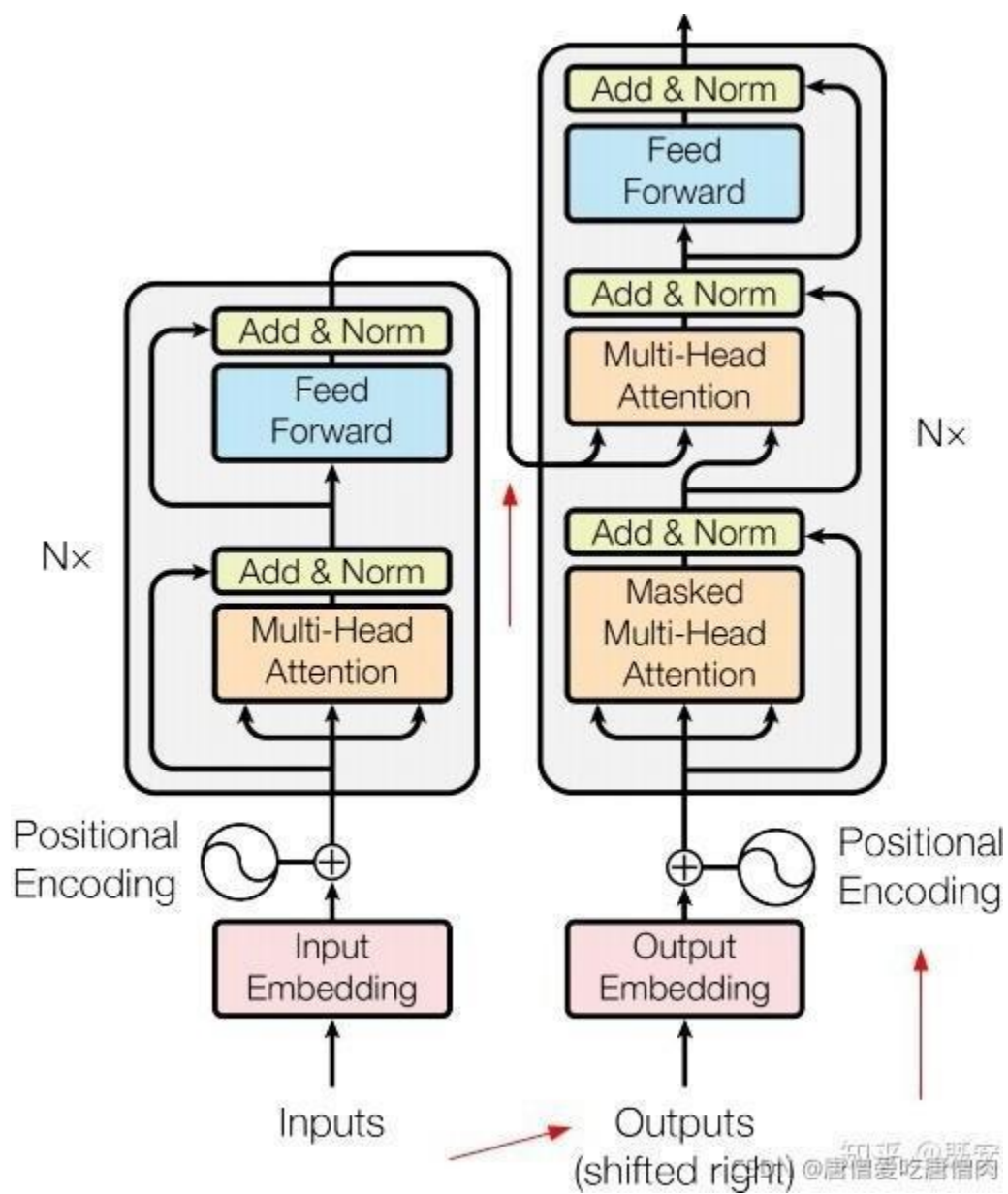
- Chatbots
- Text generation
- Question answering
- Content generation
- Code generation
- Summarization
- Creative writing

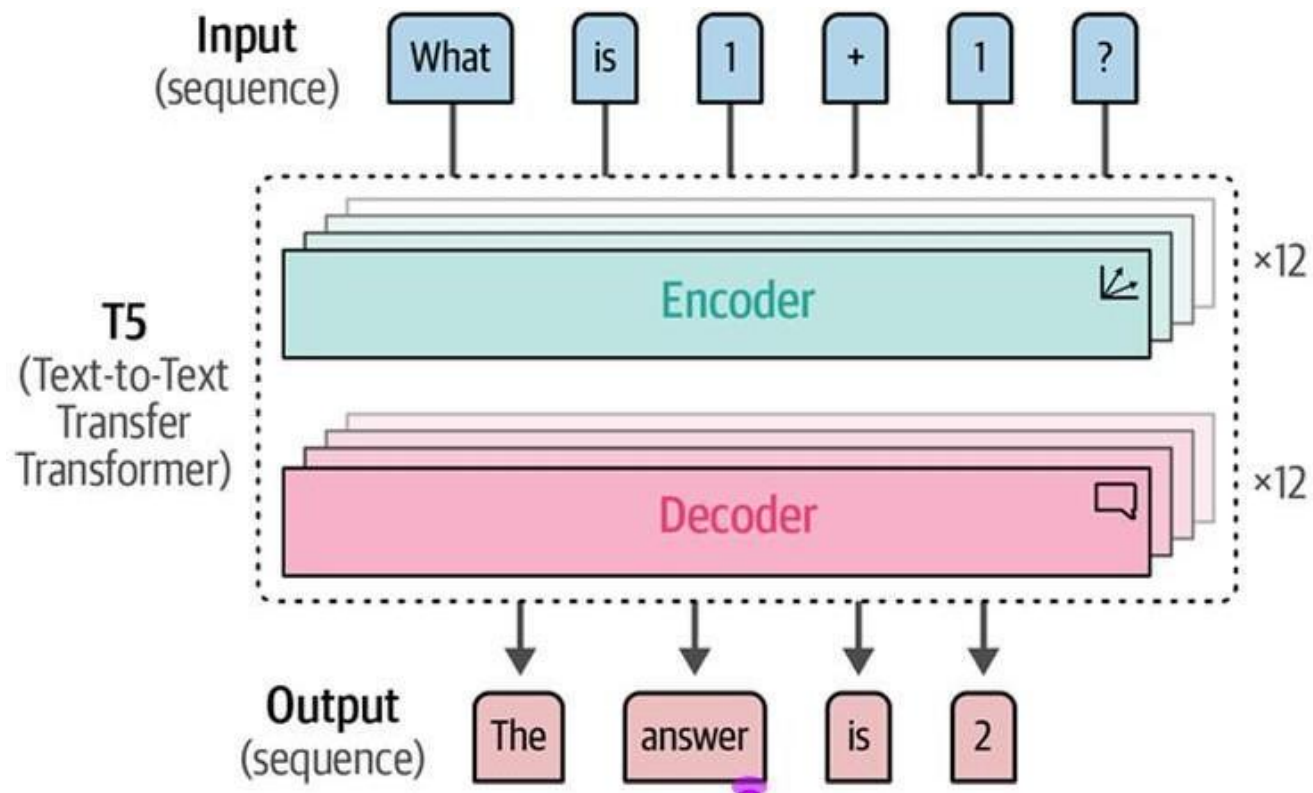
Main strength

Excellent **generative capability**.

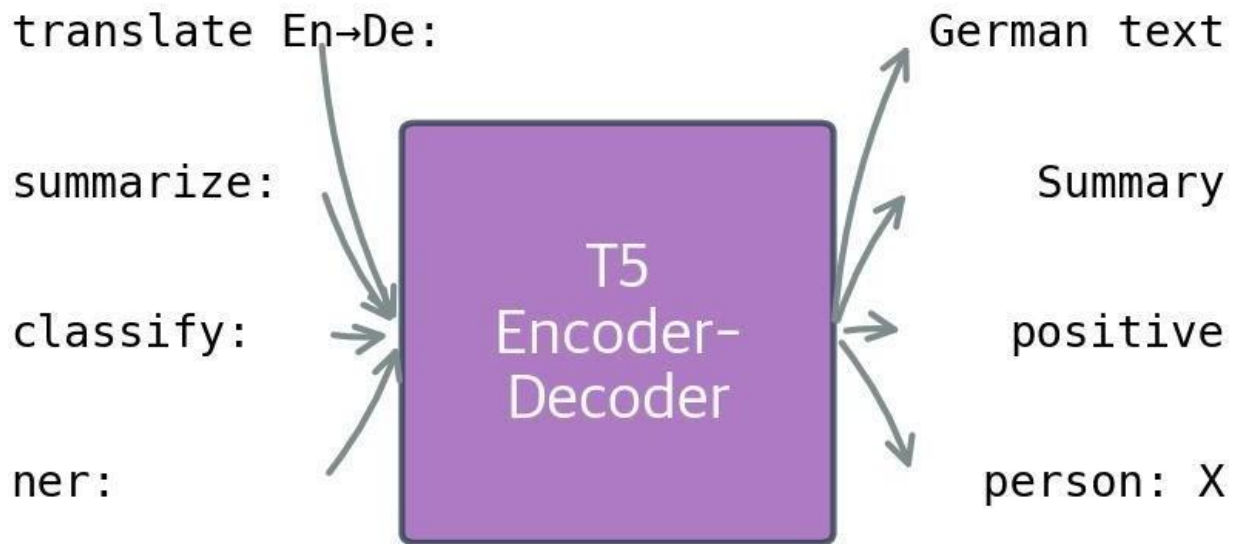
Slide 7 – T5



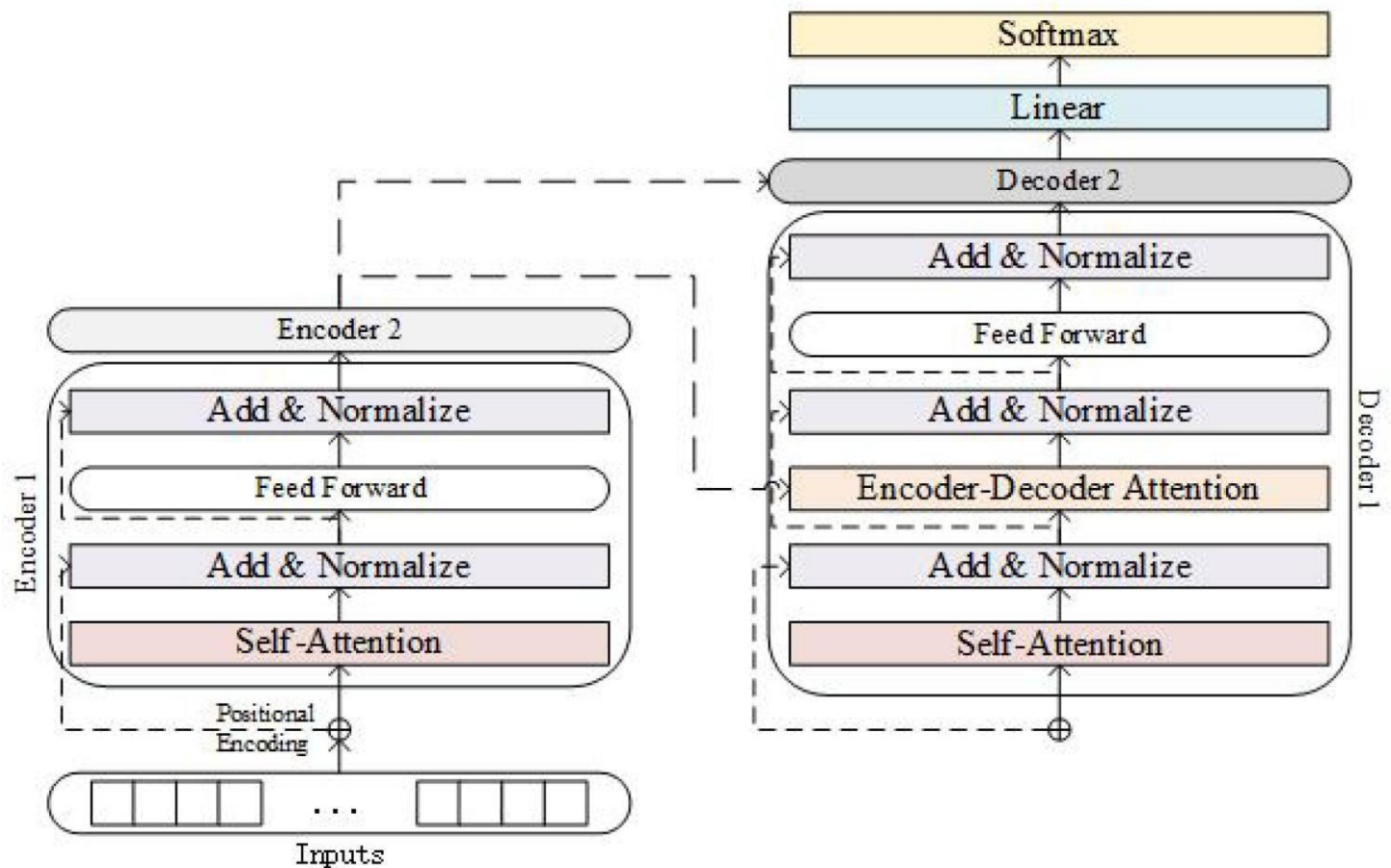




T5 Text-to-Text Approach



1 unified model



T5 = Text-To-Text Transfer Transformer

Architecture: Encoder-decoder

T5 converts almost every NLP task into a **text-to-text problem**.

Example:

Translation

translate English to French: Hello

→ Bonjour

Summarization

Summarize: [article]

→ [summary]

Question Answering

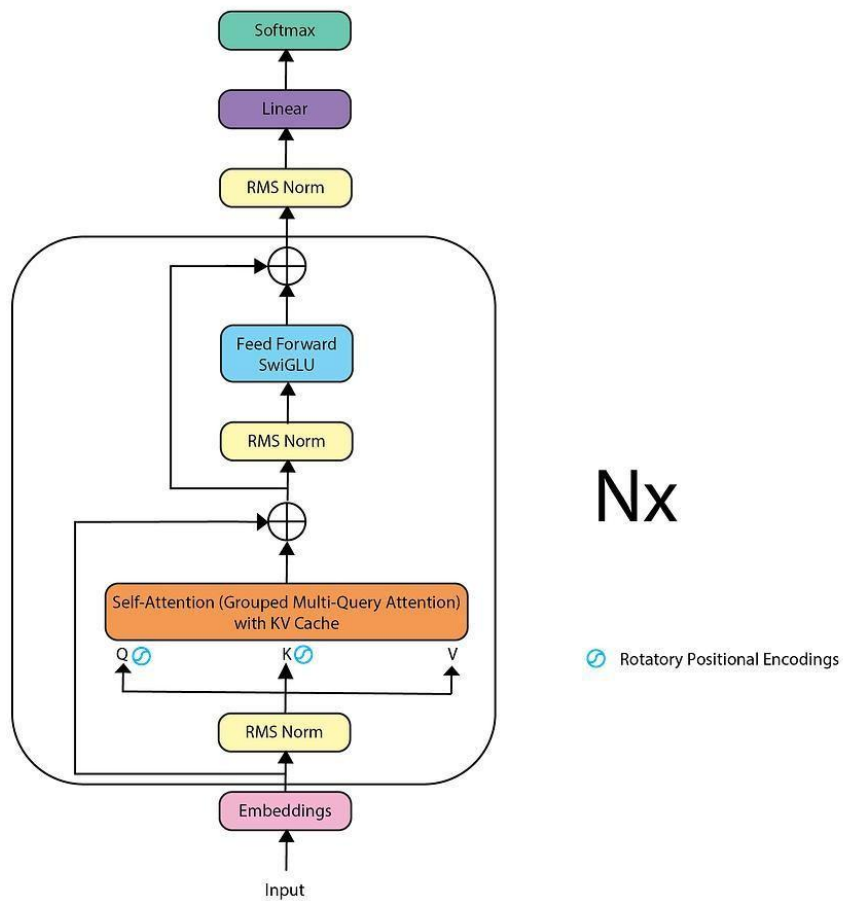
question: What is AI? context: ...

→ AI is...

Best suited for

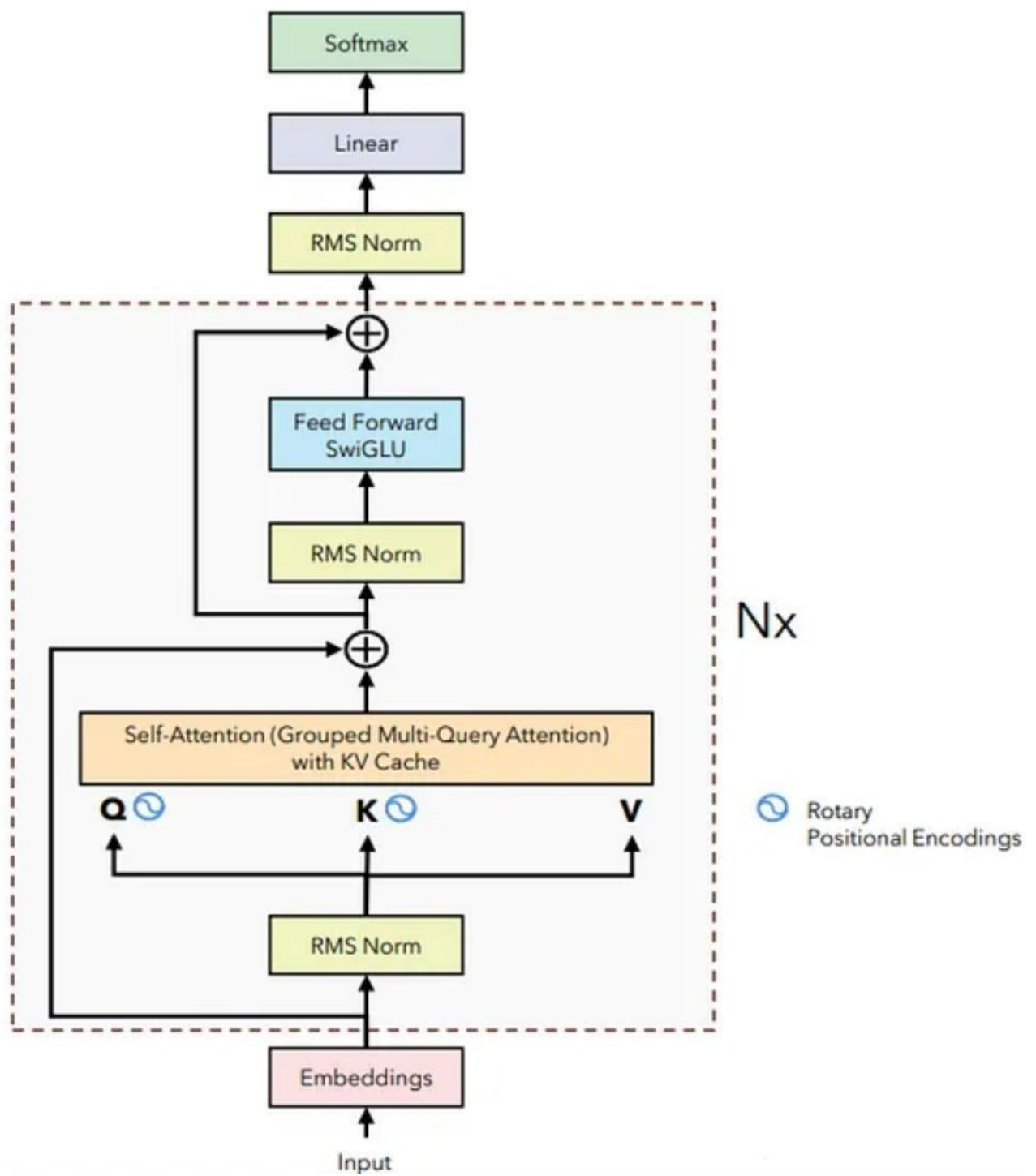
- Translation
- Summarization
- Question answering
- Text generation
- Classification
- Text transformation

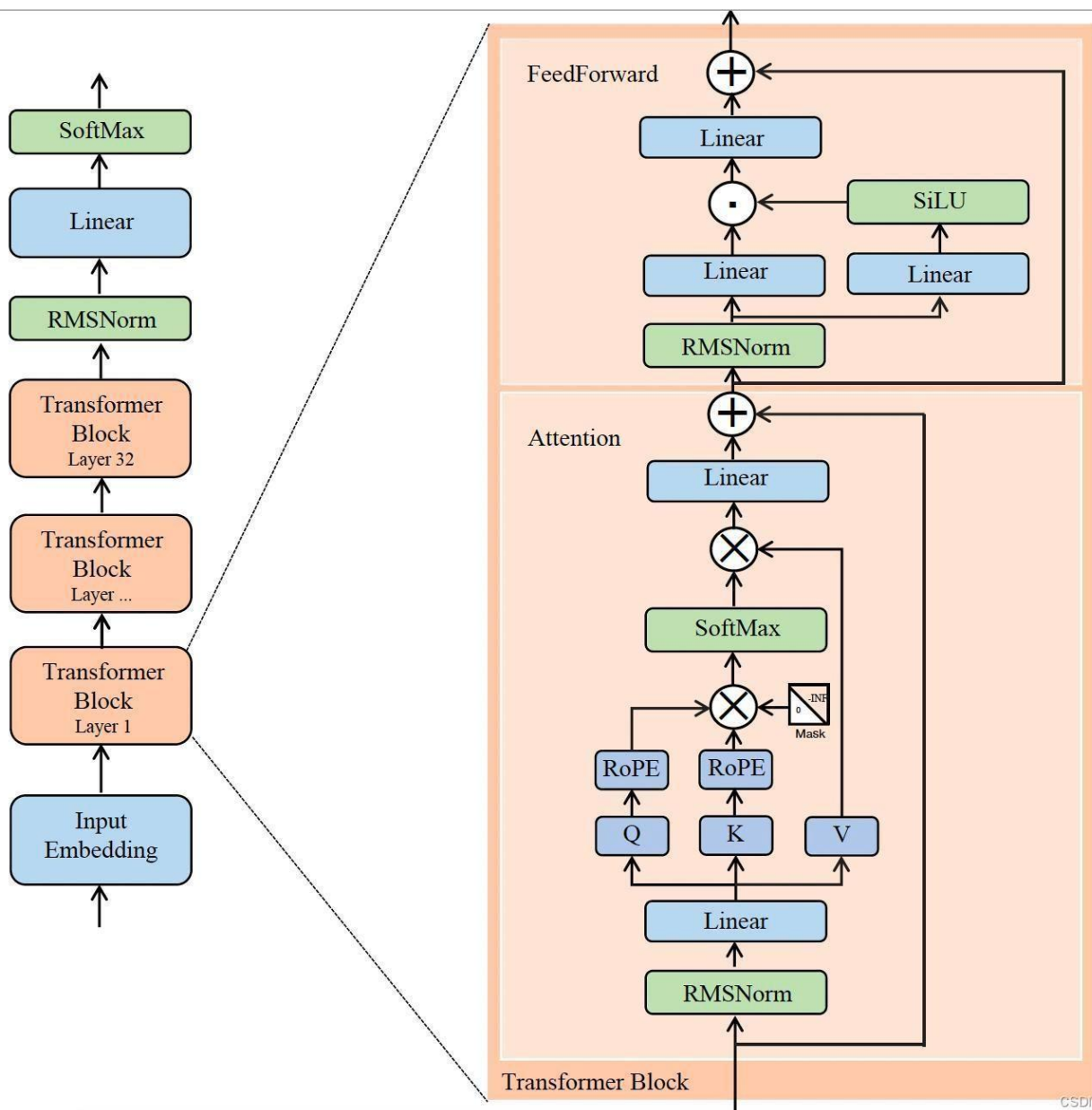
Output The quick brown fox jumps over the lazy dog [EOS]

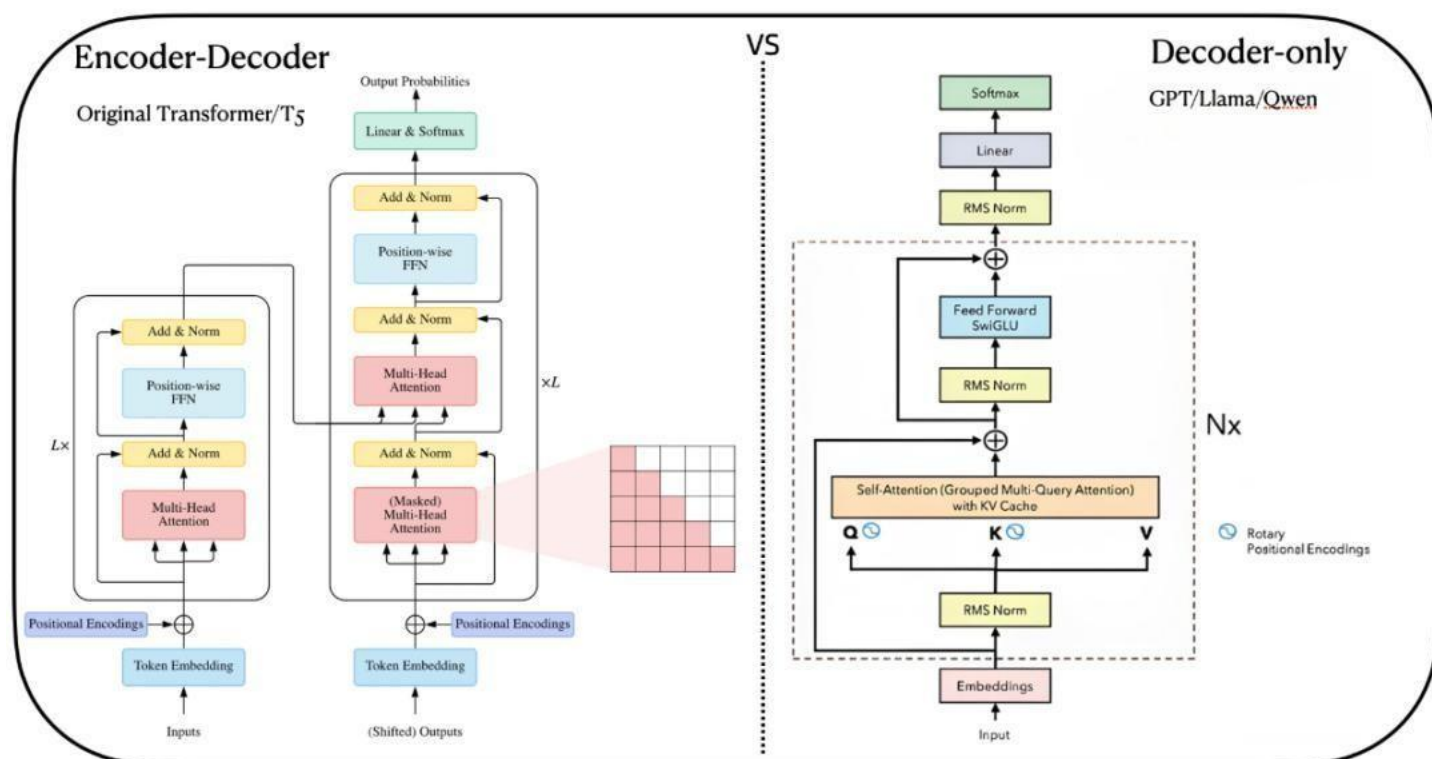


Input [SOS] The quick brown fox jumps over the lazy dog

LLaMA







LLaMA

LLaMA is a family of **decoder-only Transformer language models** developed by Meta.

Therefore:

LLaMA belongs to the same broad architectural category as GPT: decoder-only.

It uses **causal/autoregressive attention**, meaning the model predicts the next token based on previous tokens.

Best suited for

- AI assistants
- Chatbots
- Text generation
- Question answering
- Code generation
- RAG systems
- Reasoning and instruction-following applications

Major advantage

LLaMA models are available in multiple sizes and can be adapted for **self-hosted and customized applications**, subject to the particular model's license.

Slide 9 – BERT vs GPT vs T5 vs LLaMA

Feature	BERT	GPT	T5	LLaMA
Architecture	Encoder	Decoder	Encoder-Decoder	Decoder
Attention	Bidirectional	Causal	Encoder + Decoder	Causal
Main purpose	Understanding	Generation	Transformation	Generation
Text	Limited	Excellent	Excellent	Excellent

Feature	BERT	GPT	T5	LLaMA
generation				
Classification	Excellent	Good	Good	Good
Translation	Limited	Good	Excellent	Good
Summarization	Limited	Excellent	Excellent	Excellent
Chatbot	Limited	Excellent	Good	Excellent
NER	Excellent	Good	Good	Good
Code generation	Limited	Excellent	Good	Excellent

Slide 10 – Pre-Training Objectives

This is one of the **most important comparison points**.

BERT – Masked Language Modeling

Some words are hidden.

Example:

"The student is [MASK] the assignment."

BERT predicts the missing word.

This teaches BERT to understand **context from both directions**.

GPT – Causal Language Modeling

GPT predicts the next token.

Example:

"The student is"

→ "writing"

→ "the"

→ "assignment"

This makes GPT naturally suitable for **text generation**.

T5 – Text-to-Text Learning

T5 converts tasks into input-output text pairs.

Example:

Translate: I am a student

↓

Je suis étudiant

This makes T5 highly flexible across NLP tasks.

LLaMA – Autoregressive Language Modeling

LLaMA primarily learns to predict the **next token** from previous tokens.

Example:

AI is changing the

↓

world

This makes it suitable for generative applications.

Slide 11 – Input and Output Processing

BERT

Input Text



Tokenizer



Transformer Encoder



Contextual Representation



Task-Specific Head



Prediction

Example:

"This movie is excellent"



Positive

GPT

Prompt



Tokenizer



Decoder



Next Token



Next Token



Generated Text

T5

Input Text



Encoder



Encoded Representation



Decoder



Output Text

LLaMA

Prompt



Tokenizer



Decoder



Next Token Prediction



Generated Response

Slide 12 – Attention and Context

Model	Context Understanding	Attention
BERT	Strong bidirectional understanding	Bidirectional self-attention
GPT	Strong previous-token context	Causal self-attention
T5	Strong input understanding + generation	Encoder/decoder attention
LLaMA	Strong autoregressive context	Causal self-attention

Key difference

BERT:

"Understand the complete sentence."

GPT/LLaMA:

"Predict what comes next."

T5:

"Understand the input and generate an appropriate output."

Slide 13 – Application Mapping

Application	Best Choice	Reason
Sentiment Analysis	BERT	Strong classification
Text Classification	BERT	Excellent contextual representations
NER	BERT	Token-level understanding
Text Generation	GPT/LLaMA	Autoregressive generation
Chatbot	GPT/LLaMA	Natural response generation
Summarization	T5/GPT/LLaMA	Strong generation
Translation	T5	Encoder-decoder architecture
Extractive QA	BERT	Strong span extraction
Generative QA	GPT/T5/LLaMA	Text generation
Code Generation	GPT/LLaMA	Generative capability
RAG	GPT/LLaMA/T5	Generate answers from retrieved context

Slide 14 – Text Classification

BERT – Best choice

For traditional classification tasks, BERT is often an excellent choice.

Example:

Customer review:

"The product quality is excellent."

Output:

Positive

Why BERT?

- Bidirectional context
- Strong sentence representations
- Easy fine-tuning
- Relatively smaller models are available
- Good for supervised classification

GPT/LLaMA

Can perform classification using prompts:

"Classify this review as positive or negative."

But this may be less efficient than using a specialized fine-tuned encoder for large-scale classification.

Slide 15 – Summarization

T5

Very suitable because summarization naturally fits:

Long Document



Encoder



Decoder



Short Summary

GPT/LLaMA

Also excellent because they can generate summaries directly from prompts.

BERT

Not naturally designed for free-form generation.

Conclusion

T5 → strong dedicated encoder-decoder choice

GPT/LLaMA → excellent modern generative choices

Slide 16 – Machine Translation

Example:

English

"I am learning AI."



Translation Model



French

"J'apprends l'IA."

Comparison

BERT

- Excellent understanding
- Not designed for direct sequence generation
- Not the preferred architecture

GPT

- Can translate using prompting
- Strong general-purpose capability

T5

- Encoder understands source
- Decoder generates target
- Naturally suited to translation

LLaMA

- Can perform multilingual translation depending on language coverage and model capabilities
- Useful when translation is part of a broader generative application

Recommended architecture

Encoder-decoder → T5-style architecture

Slide 17 – Question Answering

There are two major types.

Extractive QA

Answer is directly extracted from the document.

Example:

Question: Where is the college located?

Document: The college is located in Chennai.

Answer: Chennai

➡ **BERT is highly suitable.**

Generative QA

The model generates a complete answer.

➡ **GPT / LLaMA / T5**

Example:

Question: Explain the admission process.

Generated answer:

"The admission process consists of application, eligibility verification, entrance requirements..."

Slide 18 – Chatbot Development

BERT

Suitable for:

- Intent classification

- FAQ matching
- Sentiment detection

Not ideal as the main conversational generator.

GPT

Excellent for:

- General-purpose chatbot
- Customer service
- Personal assistant
- Conversational AI

T5

Suitable for:

- Question answering
- Response transformation
- Summarization-based assistants

LLaMA

Excellent for:

- Custom chatbots
- Self-hosted assistants
- RAG chatbots
- Domain-specific assistants

Best choices

Cloud/API chatbot → GPT

Custom/self-hosted chatbot → LLaMA

Slide 19 – Code Generation

Model	Code Understanding	Code Generation
BERT	Good for classification/analysis	Poor
GPT	Excellent	Excellent
T5	Good	Good
LLaMA	Excellent in capable code-tuned variants	Excellent in capable code-tuned variants

Best choices

GPT / LLaMA

Examples:

- Code completion
- Bug explanation
- Code conversion
- Documentation generation
- Programming assistance

Slide 20 – Information Extraction

Applications include:

- Named Entity Recognition
- Relation Extraction
- Keyword extraction
- Event extraction

BERT

Excellent for token-level classification.

Example:

"Sundar Pichai is CEO of Google."

Sundar Pichai → PERSON

Google → ORGANIZATION

GPT/LLaMA

Can extract structured information using prompts.

T5

Can convert text into structured text.

Example:

Input:

"Apple was founded by Steve Jobs."

Output:

Founder: Steve Jobs

Company: Apple

Best traditional approach

BERT

Best generative approach

GPT/LLaMA/T5

Slide 21 – RAG Applications

RAG = Retrieval-Augmented Generation

Basic architecture:

User Question



Retriever



Relevant Documents



Context



Language Model



Answer

Role of each model

BERT

- Excellent document embedding/retrieval components

- Extractive QA
- Reranking

GPT

- Strong answer generation

T5

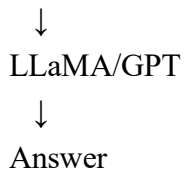
- Strong text-to-text generation

LLaMA

- Strong answer generation
- Useful for self-hosted RAG

Recommended

Retriever → BERT-based embedding/reranker



Slide 22 – Multilingual NLP

Model	Multilingual Capability
BERT	Strong with multilingual variants
GPT	Strong multilingual generation
T5	Strong multilingual variants
LLaMA	Good multilingual capability, varying by model/version

For **Indian languages**, performance can vary significantly by language, model version, tokenizer, and training data.

Languages may include:

- Hindi
- Tamil
- Telugu
- Malayalam
- Kannada
- Bengali
- Marathi

Important point

There is **no single model that is automatically best for every Indian language**.

The model should be tested on the target language and task.

Slide 23 – Model Size and Computational Requirements

A useful conceptual comparison is:

BERT

Generally smaller and efficient for classification tasks.

Advantages:

- Lower inference cost

- Easier fine-tuning
- Suitable for smaller systems

GPT

Can range from relatively small to extremely large models.

Advantages:

- Powerful generation

Disadvantage:

- Large models can require significant compute.

T5

Available in multiple sizes.

Advantages:

- Flexible
- Encoder-decoder

Disadvantage:

- Encoder + decoder computation can be expensive.

LLaMA

Available in multiple model sizes.

Advantages:

- Smaller variants can be deployed locally
- Larger variants provide stronger capabilities

Slide 24 – Fine-Tuning and Transfer Learning

BERT

Very easy to fine-tune for:

- Classification
- NER
- QA
- Sentiment analysis

GPT

Can be adapted through:

- Prompting
- Fine-tuning where supported
- Parameter-efficient methods
- Instruction tuning

T5

Highly suitable for fine-tuning because many tasks naturally fit its text-to-text format.

LLaMA

Can be customized through:

- Fine-tuning
- Instruction tuning
- LoRA
- QLoRA
- Other parameter-efficient methods

Best for customization

BERT → task-specific understanding

T5 → text-to-text tasks

LLaMA → customized generative AI

Slide 25 – Resource-Constrained Environment

For a college project or startup with limited hardware:

BERT

Best for lightweight NLP classification

Small T5

Useful for text transformation.

Small LLaMA

Can be useful for local generative AI, especially with quantization.

GPT

API-based usage can avoid local GPU requirements, but introduces service/API costs and dependency on the provider.

Practical choice

For a simple college application:

BERT → **classification**

For a generative assistant:

Small LLaMA → **local deployment**

Slide 26 – Zero-Shot, One-Shot and Few-Shot Learning

Zero-shot

No examples are provided.

Classify:

"This product is amazing."

Output:

Positive

One-shot

One example is given.

Example:

"Excellent product" → Positive

Input:

"Very useful product" → ?

Few-shot

Several examples are given.

GPT / LLaMA

Generally very strong for prompt-based few-shot learning.

BERT

Usually relies more heavily on task-specific fine-tuning.

T5

Can perform prompted/text-to-text tasks effectively.

Slide 27 – Long Context

Long-context capability is important for:

- Large documents
- Research papers
- Legal documents
- Academic regulations
- Books
- RAG

BERT

Traditional BERT has a relatively limited context window.

GPT

Modern GPT systems can support very large contexts, depending on the specific model.

T5

Context length depends on the specific T5 variant and implementation.

LLaMA

Context length varies significantly across model generations and variants.

Important

Do not compare only the model family name.

Actual context capacity depends on the specific version.

Slide 28 – Open Source and Customization

Feature	BERT	GPT	T5	LLaMA
Open model availability	Yes	Varies by model	Yes	Yes, subject to license
Local deployment	Yes	Depends on model	Yes	Yes
Fine-tuning	Excellent	Available depending on model/service	Excellent	Excellent
Prompting	Moderate	Excellent	Excellent	Excellent
Customization	High	Depends on access	High	High

Developer perspective

BERT/T5/LLaMA provide strong options for developers who need model-level customization.

GPT is particularly convenient when using an API/service rather than managing model infrastructure yourself.

Slide 29 – Strengths and Limitations

Model	Strengths	Limitations
BERT	Excellent understanding, classification	Not designed for open-ended generation

Model	Strengths	Limitations
GPT	Excellent generation, reasoning, dialogue	Large models can be expensive; model access depends on provider
T5	Flexible text-to-text framework	Encoder-decoder computation can be heavier
LLaMA	Strong generation, customization, local deployment options	Requires compute for larger models; capabilities vary by version

Slide 30 – Overall Comparison

Category	BERT	GPT	T5	LLaMA
Architecture	Encoder	Decoder	Encoder-Decoder	Decoder
Main Strength	Understanding	Generation	Transformation	Generation
Classification	★★★★★	★★★★★	★★★★★	★★★★★
Generation	★★	★★★★★	★★★★★	★★★★★
Translation	★★	★★★★★	★★★★★	★★★★★
Summarization	★★	★★★★★	★★★★★	★★★★★
Chatbot	★★	★★★★★	★★★★★	★★★★★
NER	★★★★★	★★★★★	★★★★★	★★★★★
Code	★★	★★★★★	★★★★★	★★★★★
Customization	★★★★★	★★★★	★★★★★	★★★★★

The star ratings are qualitative, not benchmark scores.

Slide 31 – Real-World Case Study: University AI Assistant Problem

A university wants an AI student-support system for:

- Admission questions
- Course information
- Regulations
- Fees
- Timetable
- Academic documents

BERT

Can classify user intent:

"How much is the tuition fee?"



Intent = Fee Query

GPT

Can generate conversational answers.

T5

Can transform retrieved information into concise answers.

LLaMA

Can provide a customizable self-hosted generative assistant.

Recommended architecture

Student



Chat Interface



Retriever



College Documents



LLaMA / GPT



Answer

Best overall: A RAG-based generative model such as GPT or an appropriately sized LLaMA, with an encoder/retrieval component where useful.

Slide 32 – Case Study: News Summarization

Requirement

Thousands of articles need automatic summaries.

BERT

Useful for:

- Article classification
- Topic detection
- Extracting important information

T5

Excellent for:

Article → Summary

GPT/LLaMA

Excellent for:

- Summarization
- Different summary styles
- User-controlled length
- Interactive summarization

Recommendation

T5 for a dedicated text-to-text pipeline

or

GPT/LLaMA for a broader generative system.

Slide 33 – Case Study: AI Coding Assistant

Requirements:

- Code completion
- Code explanation
- Debugging
- Code generation
- Documentation

BERT

Good for understanding/classification but not ideal for code generation.

T5

Can handle code-to-text and text-to-code transformations.

GPT

Excellent general-purpose code generation.

LLaMA

Strong option when using an appropriate code-capable variant and when customization/local deployment are important.

Recommendation

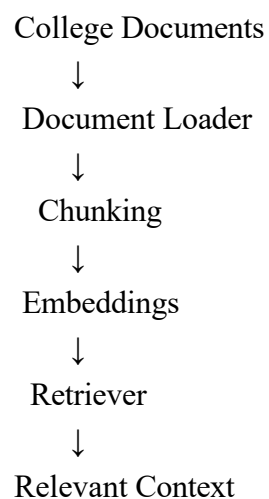
GPT or a code-capable LLaMA-family model

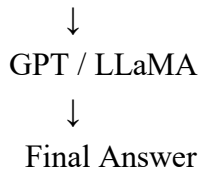
Slide 34 – Case Study: College RAG System

Suppose the college has:

- Regulations
- Syllabus
- Course documents
- Examination rules
- Attendance policies

Architecture:





Model roles

BERT: retrieval/reranking/QA

T5: text transformation/generation

GPT: answer generation

LLaMA: customizable answer generation

Recommendation

RAG + GPT or LLaMA

Slide 35 – E-Commerce Case Study

Requirements:

1. Sentiment analysis
2. Review summarization
3. Customer chatbot

Best combination

Task	Recommended Model
Sentiment Analysis	BERT
Review Summarization	T5 / GPT / LLaMA
Customer Chatbot	GPT / LLaMA

Instead of using one model for everything, a **multi-model architecture** may be more efficient.

Slide 36 – Limited-Resource Startup

Suppose a startup has:

- Limited GPU
- Small development team
- Limited budget

Option 1

Use **BERT** for classification.

Option 2

Use a **small LLaMA-family model** with quantization for local generation.

Option 3

Use a hosted GPT service to avoid maintaining GPU infrastructure.

Decision

The choice depends on:

- Hardware
- Budget
- Privacy requirements
- Latency
- Required accuracy

- Need for customization

Slide 37 – General-Purpose Generative AI Application

The company requires:

- Text generation
- Summarization
- Question answering
- Translation
- Chatbot
- Code assistance

BERT

+ Not suitable as the main general-purpose generator.

T5

■ Very flexible text-to-text architecture.

GPT

■ Excellent general-purpose generative capability.

LLaMA

■ Excellent option for customizable generative applications.

Recommended

For a general-purpose application:

GPT or LLaMA as the primary generative model

with specialized models such as **BERT** where classification/retrieval tasks benefit from them.

Slide 38 – Which Model Should a College Student Choose?

For a **college-level AI application**, there is no single answer.

If project = Sentiment Analysis

⇒ **BERT**

If project = Translation

⇒ **T5**

If project = Chatbot

■ **GPT or LLaMA**

If project = RAG chatbot

⇒ **LLaMA/GPT + retrieval system**

If project = Text classification

⇒ **BERT**

If project = General Generative AI

⇒ **GPT or LLaMA**

Slide 39 – Final Decision Matrix

Requirement	Best Choice
Text Classification	BERT

Requirement	Best Choice
Sentiment Analysis	BERT
NER	BERT
Extractive QA	BERT
Translation	T5
Text Transformation	T5
Summarization	T5 / GPT / LLaMA
Chatbot	GPT / LLaMA
Text Generation	GPT / LLaMA
Code Generation	GPT / LLaMA
RAG Generation	GPT / LLaMA
Local Generative AI	LLaMA
General-Purpose AI	GPT / LLaMA

Slide 40 – Conclusion

BERT

Best for understanding text

Encoder-only → Classification, NER, Extractive QA

GPT

Best for general text generation

Decoder-only → Chatbots, generation, code, QA

T5

Best for text-to-text transformation

Encoder-decoder → Translation, summarization, QA

LLaMA

Best for customizable generative AI

Decoder-only → Chatbots, RAG, generation, code

Final takeaway

BERT = Understand

GPT = Generate

T5 = Transform

LLaMA = Generate + Customize

The most important point is that **architecture determines the natural strengths of a model**. For a real-world system, using a **combination of models** can often be better than forcing one model to perform every task.

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Architectural Comparison Accuracy	30%	Accurate, in-depth comparison of architectures and use-cases	Accurate comparison with minor gaps	Superficial comparison of models	Inaccurate or confused comparison
Application Mapping	25%	Clearly maps each model to appropriate real-world applications	Reasonable mapping to applications	Weak mapping to applications	No meaningful mapping presented
Research Depth	20%	Well-researched with credible sources/benchmarks cited	Adequately researched	Limited research depth	Unresearched / opinion-based content
Delivery	15%	Confident, engaging, well-paced delivery	Clear delivery with minor pacing issues	Hesitant delivery	Unclear or unprepared delivery
Slide Design & References	10%	Well-designed slides with proper references	Adequate slides, minor reference gaps	Basic slides, poor referencing	No slides or references provided

Marks Evaluation:

Question No.	Architectural Comparison Accuracy (30%)	Application Mapping (25%)	Research Depth (20%)	Delivery (15%)	Slide Design & References (10%)	Total Marks (100)
Q1						
Q2						
Overall Total (100)						